

Introduction to neural networks

Warren McCulloch & Walter Pitts Model

Juan Irving Vásquez
jivg.org

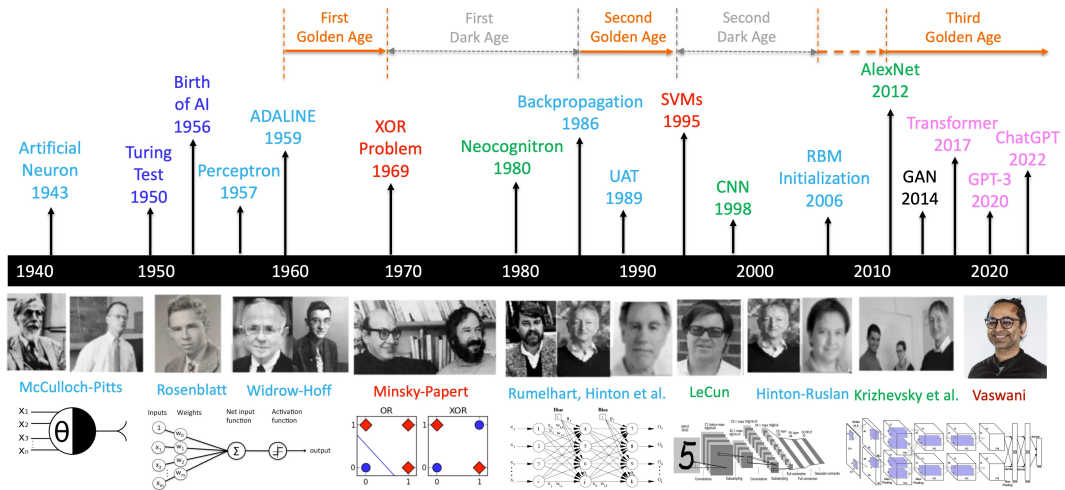
Centro de Innovación y Desarrollo Tecnológico en Cómputo
Instituto Politécnico Nacional

February 9, 2026©



1 Warren McCulloch & Walter Pitts Model

The big picture ¹



¹<https://medium.com/@lmpo/a-brief-history-of-ai-with-deep-learning-26f7948bc87b>

The problem

- Conexionism: replicate the human intelligence by simulating the brain functioning.
- The base of a brain: the neuron.
- Given an input $X \in \mathbb{B}^n$, build a machine, f , that check if the input is similar to a target signal, namely.

$$y = f(X)? \quad (1)$$

Warren McCulloch & Walter Pitts

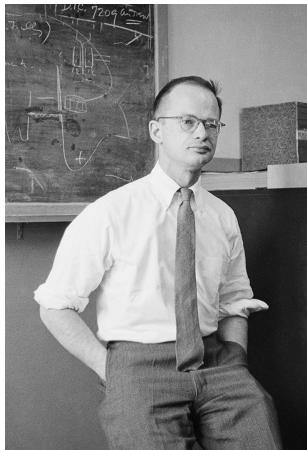
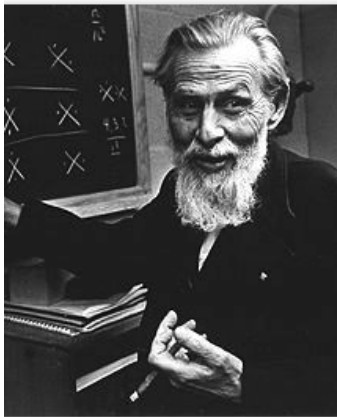


Figure: Warren McCulloch and Walter Pitts²

²https://es.wikipedia.org/wiki/Walter_Pitts

A little bit of history

- Warren Sturgis McCulloch (1898-1969) ³
 - He earned his bachelor's degree in psychology from Yale University in 1921
 - He joined the University of Illinois Chicago in 1941
- Walter Harry Pitts (1923-1969) ⁴
 - Assisted to the university of Chicago
 - At the age of 15, he had read the Principia Mathematica
 - He moved to the MIT

³<https://archives.library.illinois.edu/the-cybernetics-thought-collective-a-history-of-science-and-technology-portal-project/warren-s-mcculloch/>

⁴Smalheiser, Neil R. "Walter pitts." Perspectives in Biology and Medicine 43.2 (2000): 217-226. 

The McCulloch-Pitts model ⁵

A neuron can be modeled as function

$$\hat{y} = C(E, I, u) \quad (2)$$

where excitatory input $E = \{e_1, \dots, e_n\}$, $e \in \{0, 1\}$, $I = \{i_1, \dots, i_m\}$, $i \in \{0, 1\}$ and u is a threshold.

⁵Minsky, Marvin Lee. Computation. Chapter 3. Englewood Cliffs: Prentice-Hall, 1967.

Diagram

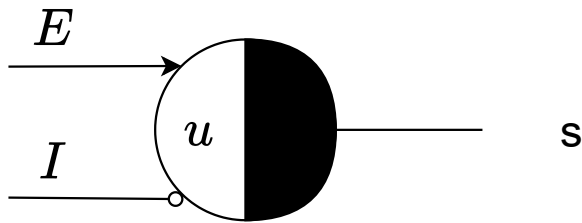


Figure: MyP graphical representation.

Data: $E = \{e_1, \dots, e_n\}$, $I = \{\iota_1, \dots, \iota_m\}$, u

Result: $\hat{y}$

```
1 if  $\exists i_j = 1; i \in I$  then
2   | return 0
3 else
4   |  $s = \sum_{i=1}^n e_i$  if  $s \geq u$  then
5     | return 1
6   | else
7     | return 0
8   | end
9 end
```

Algorithm 1: Algoritmo de inferencia de una neurona de McCulloch y Pitts.

- 1 Dibuje el diagrama de la neurona $C = (E = \emptyset, I = \{\iota_1\}, u = 0)$.
- 2 Calcule la tabla de salidas de una neurona de MyP definida por las entradas $E = [e_1]$, $I = [\iota_1]$ y un umbral $u = 1$.

- The MyP model is considered as a linear thresholding gate (LTG).
- A LTG belongs to the Thresholding gates group.
- Capabilities of the LTG:

LTG capability

Any boolean function of n variables can be realized using a two-layer network with at most 2^{n-1} AND gates in the first (hidden layer) and a single OR gate in the second (output) layer.

⁶Hassoun, Mohamad H. Fundamentals of artificial neural networks. MIT press, 1995.

- Obligatorias:
 - Carroll, L. (2024). Las aventuras de Alicia en el país de las maravillas. **capítulo 6: Cerdo y pimienta**. Chibalete Editores.
- Adicionales:
 - Minsky, Marvin Lee. Computation. Chapter 3: NEURAL NETWORKS. AUTOMATA MADE UP OF PARTS, Englewood Cliffs: Prentice-Hall, 1967.

- Minsky, Marvin Lee. *Computation*. Englewood Cliffs: Prentice-Hall, 1967.
- Juan Irving Vasquez, *Introducción a las redes neuronales*.
https://github.com/irvingvasquez/libro_redes_neuronales
- Minsky, Marvin Lee. *Computation*, Englewood Cliffs: Prentice-Hall, 1967.
- Image: <https://ib.bioninja.com.au/standard-level/topic-6-human-physiology/65-neurons-and-synapses/neurons.html>